Ttile: Modeling temporal dependence in a sequence of spatial random partitions driven by spanning tree: an application to mosquito-borne diseases

Link: http://arxiv.org/abs/2501.04601v1

Abstract: Spatially constrained clustering is an important field of research, particularly when it involves changes over time . We introduce a Bayesian model for time-dependent sequences of spatial random partitions .

Ttile: Predicting high dengue incidence in municipalities of Brazil using path signatures

Link: http://arxiv.org/abs/2501.12395v1

Abstract: In the epidemiology of dengue, environmental conditions can significantly impact the transmission of the virus . Utilizing epidemiological indicators alongside environmental variables can enhance predictions . This study analyzed a dataset of weekly case numbers, temperature, and humidity across Brazilian municipalities to forecast the risk of high dengue incidence usi .

Ttile: Mosqlimate: a platform to providing automatable access to data and forecasting models for arbovirus disease

Link: http://arxiv.org/abs/2410.18945v1

Abstract: Dengue is a climate-sensitive mosquito-borne disease with a complex transmission dynamic . Different datasets and methodologies have been applied to build complex models for dengue forecast, stressing the need to evaluate these models and their relative accuracy grounded on a reproducible methodology .

Ttile: Large-scale Epidemiological modeling: Scanning for Mosquito-Borne Diseases Spatio-temporal Patterns in Brazil

Link: http://arxiv.org/abs/2407.21286v1

Abstract: This study presents an unprecedented analysis, in terms of breadth, estimating the SIR transmission parameters from incidence data in all 5,570 municipalities in Brazil over 14 years

(2010-2023) .

Ttile: Integrating socioeconomic and geographic data to enhance infectious disease prediction in Brazilian cities

Link: http://arxiv.org/abs/2405.01422v2

Abstract: Supervised machine learning models and public surveillance data have been employed for infectious disease forecasting in many settings . These models leverage various data sources capturing drivers of disease spread .